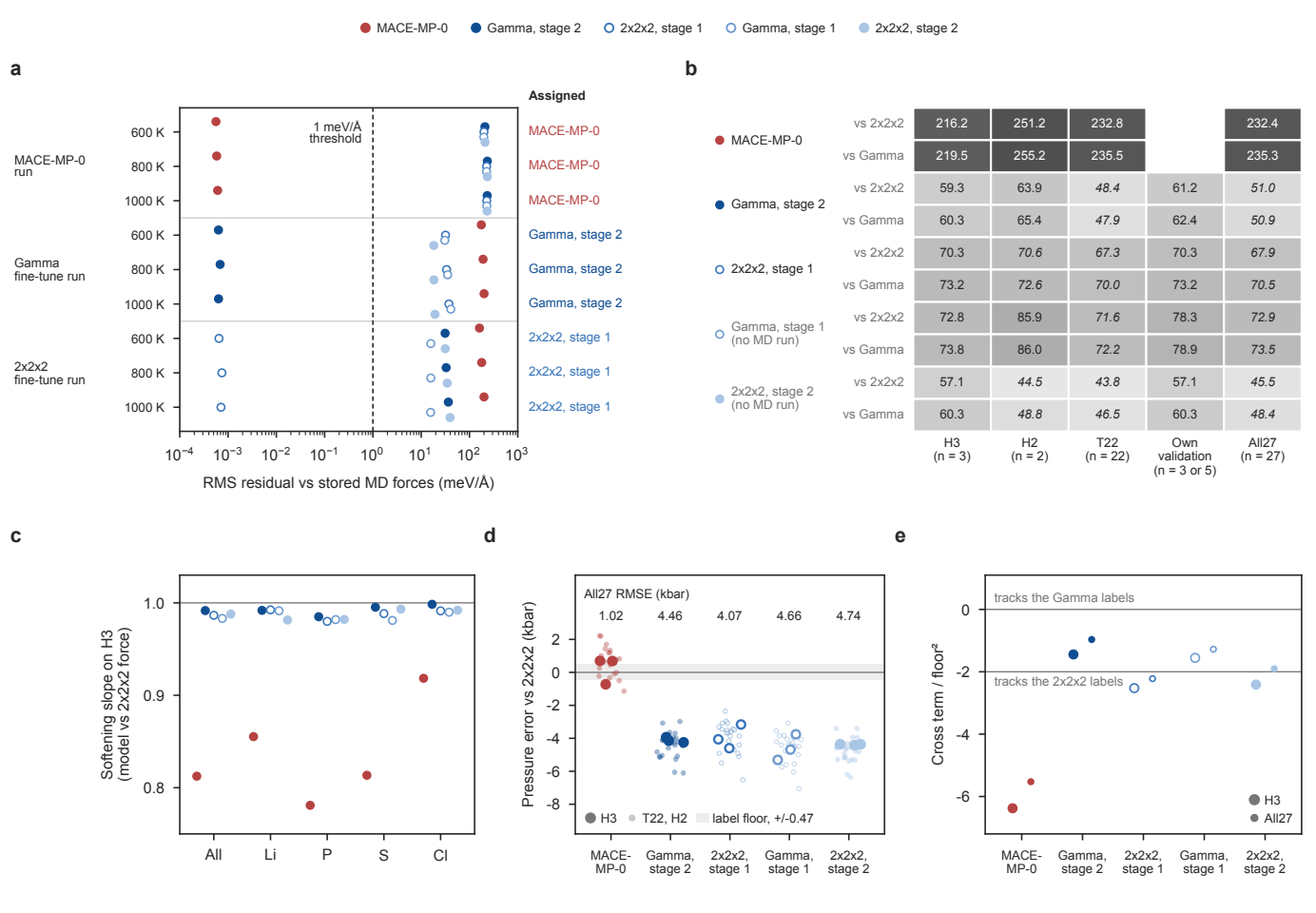

MACE-MP-0
Gamma, stage 2
2x2x2, stage 1
Gamma, stage 1
2x2x2, stage 2
a
Assigned
MACE-MP-0 run
Gamma fine-tune run
2x2x2 fine-tune run
600 K
800 K
1000 K
1 meV/Å threshold
MACE-MP-0
Gamma, stage 2
2x2x2, stage 1
RMS residual vs stored MD forces (meV/Å)
b
MACE-MP-0
Gamma, stage 2
2x2x2, stage 1
Gamma, stage 1 (no MD run)
2x2x2, stage 2 (no MD run)
vs 2x2x2
vs Gamma
216.2
251.2
232.8
232.4
219.5
255.2
235.5
235.3
59.3
63.9
48.4
61.2
51.0
60.3
65.4
47.9
62.4
50.9
70.3
70.6
67.3
70.3
67.9
73.2
72.6
70.0
73.2
70.5
72.8
85.9
71.6
78.3
72.9
73.8
86.0
72.2
78.9
73.5
57.1
44.5
43.8
57.1
45.5
60.3
48.8
46.5
60.3
48.4
H3 (n = 3)
H2 (n = 2)
T22 (n = 22)
Own validation (n = 3 or 5)
All27 (n = 27)
c
Softening slope on H3 (model vs 2x2x2 force)
1.0
0.9
0.8
All
Li
P
S
Cl
d
All27 RMSE (kbar)
1.02
4.46
4.07
4.66
4.74
Pressure error vs 2x2x2 (kbar)
2
0
-2
-4
-6
-8
H3
T22, H2
label floor, +/-0.47
MACE-MP-0
Gamma, stage 2
2x2x2, stage 1
Gamma, stage 1
2x2x2, stage 2
e
tracks the Gamma labels
tracks the 2x2x2 labels
Cross term / floor²
0
-2
-4
-6
H3
All27
MACE-MP-0
Gamma, stage 2
2x2x2, stage 1
Gamma, stage 1
2x2x2, stage 2